

Professional Engineering Practice

Washington Accord Graduate Requirements:

- Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings.
- Comprehend and apply advanced knowledge of the widely-applied principles underpinning good practice.

Professional practice module – topics

1. Course introduction & professionals
2. Law
3. Intellectual property
4. Marketing for engineers
5. Professional engineering practice (today)
 - Professional engineering
 - Professional meetings
 - Personal time management
 - AI in professional practice

Engineering Industry

- Four main sectors within industry:
 - Asset owners
 - Consultancies
 - Contractors
 - Manufacturers & product designers
- Each have different operational, work structures

Engineering Industry – Asset owners

Orion



Synlait

Te Whatu Ora
Health New Zealand



Engineering Industry – Consultancies



Engineering Industry – Contractors



TELTRAC

*Advanced
Security*



Engineering Industry – Manufacturers & designers



Kea Aerospace



Rockwell
Automation



HamiltonJet



Fisher & Paykel
HEALTHCARE



enatel

energylight
knowledge | solutions



Project Procurement

- Ways to obtain projects include:
 - Request for proposals (RFP)
 - Existing/historical clients
 - Networks & Word of mouth
 - Social media
- May take a long time to progress from contact to contract



ONLINE SERVICES

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POLICIES & INFORMATION

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Welcome to the New Zealand Government Electronic Tenders Service (GETS)

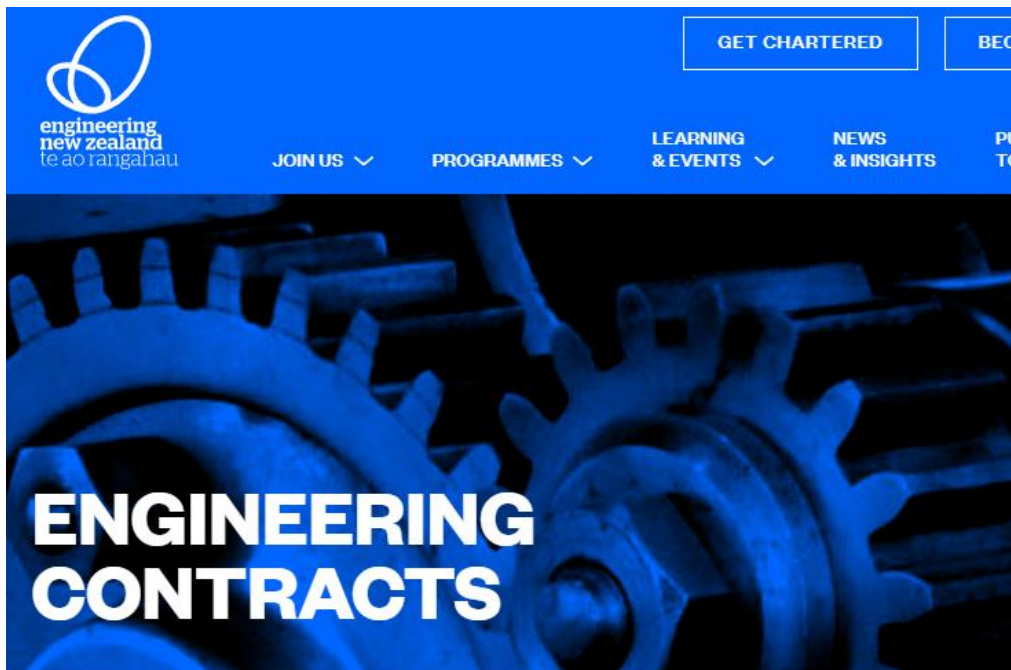
GETS is a free service designed to promote open, fair competition for New Zealand Government contract opportunities. [More...](#)

Current tenders

RFX ID	Reference #	Title	Tender Type	Close Date ▲	Organisation
29845557	24-374	B2466 - Humidifier Replacement Project	RFO	2:00 PM 21 Aug 2024 (Pacific/Auckland UTC+12:00)	University of Canterbury
29798907	11238	Retaining Wall Construction 2024/25	RFT	4:00 PM 21 Aug 2024 (Pacific/Auckland UTC+12:00)	Dunedin City Council
29563096	[None]	Data Analytics and Information Management- RFI	RFI	5:00 PM 21 Aug 2024 (Pacific/Auckland UTC+12:00)	New Zealand Defence Force
29797031	TKKM Kokohuia	Main Building Contractor for AMS Alterations & Refurbishment to Blocks 6 & 10 at TKKM Kokohuia, Whanganui	RFT	5:00 PM 21 Aug 2024 (Pacific/Auckland UTC+12:00)	Ministry of Education - School Infrastructure
29767581	MOE25930	Newlands College Block AL Weathertightness Remediation Works- Main Contractor	RFT	5:00 PM 21 Aug 2024 (Pacific/Auckland UTC+12:00)	Ministry of Education - School Infrastructure
29902040	215513	Te Akau ki Pāpāmoa School - Refurbishment of B Block	ROI	5:00 PM 21 Aug 2024 (Pacific/Auckland UTC+12:00)	Ministry of Education - School Infrastructure
29838519	2024-C502	2024-C502 - Playground Renewals FY24/25	RFP	2:30 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Kapiti Coast District Council
29889303	EA00311	Comparison and Switching Website	ROI	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Electricity Authority
29802534	CC10050	Foster Park - Youth Zone and Central Space	RFP	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Selwyn District Council
29799735	114.00315	Grosvenor Terrace Retaining Walls Seismic Strengthening	RFT	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Wellington City Council
29950204	24-452	Hollow Cylinder/Torsional Shear Soil Element Test Apparatus	NOI	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	University of Canterbury
29789430	CC10049	Lincoln Reserve Extension - Consultancy Services	RFP	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Selwyn District Council
29897398	NDOCS-224730439-87657	Nelson City Council - Paru Paru Pumpstation Upgrade Construction Contractor Pre-Construction Services (ECI) - ROI - 1Aug2024	ROI	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Nelson City Council
29810077	C00246.NSK.300	Walter Mildenhall Park - Hard and soft landscape works	ROI	4:00 PM 22 Aug 2024 (Pacific/Auckland UTC+12:00)	Hutt City Council

Contracts – liability & insurance

- **All** contracts you participate in should have a limited liability clause
 - Limits the amount owed in event of claims, damages, torts etc.
 - Some clients require this (plus matching insurance) before signing
- Should be backed with:
 - Professional indemnity insurance
 - Public liability insurance
 - Statutory liability insurance (if appropriate)



We encourage all Engineering New Zealand members to follow recognised professional practice when communicating with clients on commercial matters. To help you do this, we worked with ACE New Zealand and other key stakeholders to develop two standard contractual documents.

These are the latest versions, developed in 2017. These documents are available to members to download and use free of charge. They are subject to copyright and must not be altered.

6. Liability and Insurance

6.1 Liability

Where the Consultant breaches this Agreement, the Consultant is liable to the Client for reasonably foreseeable claims, damages, liabilities (including any liability of the Client to a third party), losses or expenses caused directly by the breach.

A Party shall not be liable to the other Party under this Agreement for the other Party's indirect, consequential or special loss, or loss of profit, however arising, whether under contract, in tort or otherwise.

6.2 Limitation of Liability

The maximum aggregate amount payable, whether in contract, tort or otherwise, in relation to claims, damages, liabilities, losses or expenses, is as specified in the Special Conditions.

6.3 Contributory Conduct

If either Party is found liable to the other (whether in contract, tort or otherwise), and the claiming Party and/or a Third Party has contributed to the loss or damage, the liable Party shall only be liable to the proportional extent of its own contribution.

6.4 Duration of Liability

Without limiting any defences a Party may have under the Limitation Act 2010 or any other legislation, neither party shall be liable for any loss or damage occurring after the period stated in the Special Conditions from the date on which the Services were completed.

6.5 Insurance

The Consultant shall take out and maintain for the duration of the Services:

- professional indemnity insurance for the amount of the liability under clause 6.2; and
- public liability insurance cover as set out in the Special Conditions; and
- other insurance as set out in the Special Conditions; and
- provision for reasonable defence costs.

SPECIAL CONDITIONS – PART A Specific Conditions of Contract

References from clauses in the General Conditions of Contract for Consultancy Services

Clause N°

Agreement

Additional documents forming part of the Agreement include:

.....

6.2 Limitation of Liability*

The maximum amount payable shall be five times the fee with a minimum limit of \$500,000 and a maximum limit of \$2,000,000;

OR

The maximum amount payable shall be \$..... (Delete one as appropriate)

Failed transport hub: Tauranga council appeals High Court decision over costs

6:05 pm on 8 August 2024

Alisha Evans, Local Democracy Reporter

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The site in June 2024 after construction of Panorama Towers began. Photo: Alex Cairns / Bay of Plenty Times

TCC went to court claiming there were contract breaches by contractors Harrison Grierson Holdings, which designed the structure, and Constructure Auckland Ltd, which was engaged to review the structural design.

Losses included more than \$20.5m in "wasted costs" and the loss of land value exceeding \$5.3m, the council said.

In the High Court case in May 2023, the defendants argued the liability limitation clauses in contracts capped how much they could be liable to pay.

Justice Kiri Tahana released her decision in March 2024, stating the limitation of liability clauses in the contracts did apply.

This set the maximum liability of Harrison Grierson at \$2m in the aggregate and Constructure Auckland's liability to a maximum of \$500,000.

HIGH COURT UPHOLDS SHORT FORM AGREEMENT



By Engineering New Zealand | 22 APR 2024



The standard short form agreement template from Engineering New Zealand and ACE New Zealand has recently been upheld by the High Court, reinforcing the robustness of the contract widely used in engineering and construction.

The Court found the liability clause within the contract was enforceable, and therefore limited the extent to which companies using the contract could be held liable.

This maximum liability is \$500,000 (exc. GST) in the Short Form Agreement and, in the Long Form Agreement, the maximum aggregate amount payable as specified in the Special Conditions.

Mistake-prone engineer accused of causing \$19 million in losses wasn't insured

Martin Van Beynen

July 01, 2022 - 05:00am

➦ Share



230 High St in central Christchurch is embroiled in a complex legal wrangle. The building has stood empty for four years.

MARTIN VAN BEYENEN / STUFF

The engineer who exposed Christchurch ratepayers to a \$19 million-plus liability by [designing a defective eight-storey building](#) in the city's CBD was not insured against his mistakes.

Professional Communications

Workplace Tip:

Make sure you get all agreements, instructions or important information in writing!

- Choose the correct method for your message:
 - Formal communication tool (e.g., Aconex, Jira)
 - Email
 - Internal messaging tool (e.g., Slack, MS teams)
 - Phone
 - Meeting
- Consider the purpose & audience
- Maintain professional language, behaviour

Professional Communications – methods



NEW ZEALAND / BUSINESS

'Unprofessional' Teams chat stand-downs a warning for others - lawyer

11:49 am on 16 July 2024

Share this     

An employment lawyer says workers should think twice before writing to colleagues on any platform, including [apps such as Microsoft Teams, Slack](#), or even Facebook Messenger.

Eight staff at Immigration NZ have been stood down after a series of what the agency is calling "inappropriate and unprofessional Teams chats" [were discovered while investigating an unrelated complaint](#).

Your boss can probably read your emails and messages

Employment law specialist Alastair Espie, a partner at Duncan Cotterill, said people should assume as a starting point that anything they sent on a work device or any work communication tool such as email or internal messaging would be "fair game" for the employer to access.

"You generally assume it will be viewable by the employer who has a right to access it. The only exception is if the employer has led staff to believe that it will not look at communications and will treat them as private."

He said sometimes employers had a policy to that effect or communicated it to staff, but it was not common.

Professional Communications – methods

- Email:
 - Good for informal communication
 - Consider your CC list. Emails can be forwarded – beware!
 - Generally locked to your individual account
- Phone calls
 - Great for quick conversations, building relationships
 - Generally not recorded, follow up in writing

Professional Communications – Meetings

- Good for coordination, efficient decision making
- Professional meetings have:
 - An agenda
 - Minutes
 - A chair
 - A secretary
- Not all meetings require these, consider size and context

Professional Communications – Meeting agenda

- Lists the activities in the meeting – indicates the purpose and scope
 - Indicates where, when and who
 - Absences should be communicated in advance as “apologies”
- Call for agenda items issued in advance of meeting
- Items may be scheduled (time) and/or “timeboxed”
- May contain reports from apologies in their stead

Engineering Faculty Forum - call for Agenda items



Yolande Lawrence

To: **ENG C Staff**

Cc: UCSA Engineering Representative; Mel Tainui; Chris Hyndman; Kathryn Mackinver; Tamsin Sheen

Kia ora koutou,

All Faculty of Engineering staff are invited to attend the next Engineering Faculty Forum, scheduled for Wed. 5th June, 1-2pm in the John Britten 102 Foyer.

If you would like to request an item be added to the draft [Faculty Forum Agenda](#), please send this via email to me by 10am next Wed. 29th May, and the confirmed Agenda will be circulated that afternoon.

Ngā mihi,
Yolande



Yolande Lawrence
PA to the Executive Dean, Prof. Saurabh Sinha
Faculty of Engineering | Te Kaupeka Pūhanga
T: +64 3 369 0908 (ext: 90908) or M: 021 074 1188
University of Canterbury | Te Whare Wānanga o Waitaha
Private Bag 4800, Christchurch 8140, New Zealand



Faculty of Engineering : Te Kaupeka Pūhanga

Department of Electrical and Computer Engineering: Te Tari Pūhanga Hangarau

Tel: +64 3 364 2264, www.elec.canterbury.ac.nz



Departmental Meeting Electrical and Computer Engineering Te Tari Pūhanga Hangarau

Agenda

Date:	14 May 2024
Time:	9am
Venue:	Rata 120
Apologies:	Enda Crossin,

- 1 Welcome and apologies.
- 2 Previous minutes, matters arising.
- 3 HOD report.
- 4 Updates:
 - 4.1 Technical services and Facilities: Philipp.
 - 4.2 Health, Safety and Wellbeing: Gareth.
 - 4.3 Undergraduate overall and EEE: Neville.
 - 4.4 CE Programme: Le.
 - 4.5 MT Programme: Michael.
 - 4.6 MEM: Enda.
 - 4.7 Outreach/Marketing: Martin.
 - 4.8 Research: Martin.
 - 4.9 Postgraduate: Chris.
 - 4.10 Learning & Teaching: Michael.
- 5 General business.

UC Strategic Plan pillars:

1. UC as an Engaged University
2. Internationalisation – Locally Engaged, Globally Networked
3. Education – Accessible, Flexible, Future Focused
4. Research – Impact in a Changing World
5. People – Nurturing Staff, Thriving Students
6. Organisational Efficacy
7. Environmentally Sustainable

University of Canterbury Private Bag 4800, Christchurch 8140, New Zealand. www.canterbury.ac.nz

Professional Communications – Meeting minutes

- Documents the meeting – what was discussed, what was decided?
 - Documents the date, time, venue used, attendees, apologies
 - Balanced between detailed and summary – keep it brief
 - May indicate when meeting concluded, next meeting time
- Assigns key actions and deadlines for key tasks – action items
- Minutes are sent out after the meeting, contents approved by all attendees

Departmental Meeting
Electrical and Computer Engineering
Te Tari Pūhanga Hangarau

Minutes

Date:	Wednesday, 24 July 2024
Time:	9-10am
Venue:	Rata 120
Apologies:	Martin Allen, Allan McInnes, Andrew Lapthorn, Lui Holder-Pearson, Paul Gaynor
Attendees	Alan Wood, Gareth Barwell, Chris Hann, Ciaran Moore, Daniel Morris, Enda Crossin, Hannelie Nel, Jeremy Watson, Joe Chen, Le Yang, Leigh Brook, Michael Hayes, Neville Watson, Philipp Hof, Philippa Martin, Richard Clare, Jodi Kilroe (minutes).

1 Welcome and apologies.

- Alan welcomed everyone to the meeting and apologies were noted. A very warm welcome to Hannelie Nel, who will be managing the conversion of the MEM programme to UC Online.

2 Previous minutes, matters arising.

- The minutes of the previous meeting were accepted.
- Action points from the last meeting were reviewed:
 - Chris has had several long discussions about EFTs with Lisa Carter and believes the solution will come once it moves to an online system.
 - Staff were asked to talk to students who enrol for an ME about expectations and reality around the timeframe of the ME. This is ongoing.
 - High School visits on 30th August: Alan is the only one doing a talk at present. Please see Alan if you are interested.
 - EFTS splits: Please let Deborah know if you are supervising a student in another department (ongoing).
 - Training date options for OurUC has been forwarded to Lui by Neville.
 - Alan was yet to speak with Andrew Bainbridge Smith about a reciprocal arrangement with COSC regarding attendance at the Learning and Teaching committees for our departments.
 - Thesis in Three preparations were currently underway for Chris.
 - Staff to see Shayne if they would like to work more with industry (ongoing).

Meeting ended at 9.45 am.

The next meeting will be Wednesday, 14th August in Rata 120.

❖ **Action Points:**

- High School visits on 30th August - Please see Alan if you are interested in giving a talk.
- Alan to speak with Andrew Bainbridge Smith about a reciprocal arrangement with COSC regarding attendance at the Learning and Teaching committees for our departments.
- FYP practicals - Please ensure risk assessments for those 50 volts DC and above are completed by the students in charge of their projects.
- Chris to provide an update about PhD doctoral scholarships once he has had an update from Lisa Carter about her discussions with Cheryl de la Rey.

Professional Communications – Meeting roles

- Chair:
 - Runs the meeting, keeps order
 - Introduces agenda items
 - Makes sure everyone is heard
 - This should be a role that seeks consensus rather than one that pushes an agenda
- May be a regular role, or rotated

Professional Communications – Meeting roles

- Secretary:
 - Calls for agenda items
 - Compiles agenda and circulates it in a timely fashion together with supporting documents
 - Takes note of apologies and quorum requirements
 - Takes minutes, circulates minutes to all meeting invitees, both present and absent
 - Updates relevant papers (in this case, attendance records)

Professional Communications – Meeting format

- Apologies
- Review minutes of previous meeting
 - True and accurate record?
 - Report on action items
- Business of meeting
- Other business?

Dressing for work and work stress

- Appearing professional is important in engineering
 - Professional attire isn't always suitable
 - PPE – provided by employer
- Stress exists in professional work
 - Prioritise mental wellbeing
 - It's ok to say no!
 - Work with your team, manager

Time Management

- Engineers are busy people!
- Managing your own time is your responsibility – no-one likes a laggard.
- Start paying attention to your own timings and methods now
- know thyself

Time Management Tips

- Use a calendar
- List & prioritise goals for the week
- Focus on one thing at a time
- Schedule in down time/recreation time, take breaks between tasks

Time Management – two methods



© Dr. Tanveer Hussain

Artificial Intelligence

- AI tools are increasingly used
 - Not mandatory, but you should be aware of them
- Current uses: writing text, writing software, creating images, predictive models
 - Future uses: design, testing, validation?

Artificial Intelligence – Privacy and IP

- AI tools are often “leaky”
 - Previously entered data can be extracted – data reconstruction
 - User details can be gleaned – membership inference
- Unclear who owns AI creations
 - Patents, copyright, designs etc.
 - What if an AI creation impinges on existing IP?

Journalism Fights Back – Why the New York Times' AI Lawsuit Matters



LAST UPDATED JANUARY 3, 2024 2:32 PM
BY JAMES MORALES

CCN

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The New York Times argues that OpenAI's use of its journalism amounts to copyright infringement.

KEY TAKEAWAYS

- The New York Times is suing OpenAI and Microsoft.
- The publisher argues that copyright law should prevent the unauthorized use of its articles as AI training data.
- In the last year, content creators of various stripes have brought similar litigation against OpenAI and other AI developers.

FORBES > INNOVATION > AI

AI Is Deciphering Your Corporate Trade Secrets

Neil Sahota Former Contributor @

Neil Sahota is a globally sought after speaker and business advisor.



May 22, 2024, 10:00am EDT



AI has become the ultimate corporate spy. GETTY

In the high-stakes world of corporate competition, trade secrets are the hidden gems that give companies their competitive edge. From the recipe for Coca-Cola to Google's search algorithm, these secrets are closely guarded. However, with the advent of AI, the sanctity of these corporate treasures is under threat. AI's ability to analyze, learn, and decipher complex patterns and data sets has opened a Pandora's box of potential for corporate espionage and the unauthorized unraveling of trade secrets.

Yes, GitHub's Copilot can Leak (Real) Secrets

Researchers successfully extracted valid hard-coded secrets from Copilot and CodeWhisperer, shedding light on a novel security risk associated with the proliferation of secrets.



THOMAS SEGURA

12 OCT 2023 · 6 MIN READ

Share



Your Code Secret Belongs to Me: Neural Code Completion Tools Can Memorize Hard-Coded Credentials

YIZHAN HUANG, The Chinese University of Hong Kong, China

YICHEN LI, The Chinese University of Hong Kong, China

WEIBIN WU*, Sun Yat-sen University, China

JIANPING ZHANG, The Chinese University of Hong Kong, China

MICHAEL R. LYU, The Chinese University of Hong Kong, China

Analyzing Leakage of Personally Identifiable Information in Language Models

Nils Lukas^{*§}, Ahmed Salem[†], Robert Sim[†], Shruti Tople[†], Lukas Wutschitz[†] and Santiago Zanella-Béguelin[†]

^{*}University of Waterloo, [†]Microsoft

nlukas@uwaterloo.ca, {t-salemahmed, rsim, shruti.tople, lukas.wutschitz, santiago}@microsoft.com

What Was Your Prompt? A Remote Keylogging Attack on AI Assistants

Roy Weiss, Daniel Ayzenshteyn*, Guy Amit, Yisroel Mirsky*

Ben-Gurion University, Israel

Offensive AI Research Lab[†]

Demo Video: <https://youtu.be/UfenH7xK0Is>

Artificial Intelligence – Privacy and IP

- Rule of thumb: assume all information input to an AI tool is publicly accessible!
 - Especially private information, trade secrets, personal information etc.
- Discuss AI tool use with managers **before you use them**
 - Check in regularly, keep up-to-date with developments and regulations